

Data Analytics I — Assignment 1

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The objective of this assignment is to introduce you to data visualization and develop an understanding of data processing using Python.

For this assignment, you are provided with two comprehensive datasets: one containing information on **bank term-deposit marketing calls**, and the other detailing **Airbnb short-term rental listings in New York City**.

Use visualisation techniques such as histograms, distribution plots, scatter plots, quantile plots, correlation matrices, and categorical groupings as a basis for constructing your inferences and answers. The suggested techniques listed after each question are provided for reference; you are encouraged to go beyond these and incorporate more in-depth, creative, and insightful analyses and plots where appropriate.

Part I [30 Marks]

1. Education Distribution & Grouping [5 Marks]

Identify the unique education levels in bank.csv and their frequencies. Group them into broader, meaningful categories (e.g., basic, intermediate, higher, not-reported). Explain the reasoning behind your mapping and support your answer with visual evidence.

2. Age–Contact Engagement Intensity Relationship and Grouping [10 Marks]

Examine the distributions of age and last-call duration. Create broader groups for each variable (e.g., brief / standard / extended / deep-engagement for contact intensity) and compare category counts and distributions before and after grouping. Analyse the relationship between age and contact-engagement intensity before and after grouping. Comment on whether grouping improves interpretability or reduces useful detail, supported by statistical evidence and visualisations.

3. Account Balance Composition and Group Performance [10 Marks]

Analyse the distribution of balance, both overall and for non-zero values, decomposing it into a gain-like component (positive balance) and a loss-like component (negative balance), and report the proportion of clients with zero balance ("no capital activity") versus positive versus negative balance. Using your age and contact-intensity groupings from the prior question, compare groups on average net balance and proportion with non-zero balance. Illustrate patterns using visualisations. Discuss whether age or contact intensity shows a stronger association with net balance, and how grouping impacts interpretability.

4. Final Dataset Refinement and Structure [5 Marks]

Summarise the final refined dataset structure, noting newly created, transformed, categorised, or grouped features, and any that were removed. Compare category counts before and after grouping for key variables, highlight changes in missing/unknown values, and describe how these refinements improve interpretability and readiness for modelling (e.g., predicting deposit).

Part II [60 Marks]

1. Price Segmentation & Market Overview [7.5 Marks]

Divide the listings into three **price tiers** and analyse their distribution across **boroughs**. Provide a high-level market summary for each price tier, incorporating **property/room type**, **borough**, and key **amenities** to give a would-be host or guest a quick snapshot of the market.

2. City-Level Comparative Analysis (Manhattan vs Brooklyn) [7.5 Marks]

Compare listing opportunities in **Manhattan** and **Brooklyn** by analysing differences in **room/property types**, guest capacity (accommodates, as your size proxy), and **prices**. Highlight both the residential segment and the small commercial/hotel-style segment (room_type == "Hotel room") for a complete picture.

3. Location-Based Premium Analysis [5 Marks]

Within each borough, compare high-budget **prime-location** listings (define "prime" as neighbourhoods whose median price sits in the top quartile of that borough) to similar **non-prime-location** listings in terms of average guest capacity, amenity count, and price-per-guest. Identify whether location justifies the price premium.

4. Value-for-Money Opportunities [10 Marks]

Identify listings offering the highest guest-capacity-per-dollar (accommodates / price) across all boroughs. Present borough-wise rankings and insights into which segments deliver the best value for budget-conscious guests.

5. Feature & Amenity Impact on Price [5 Marks]

Analyse how various amenities (e.g., Gym, Pool, Elevator, Hot tub, Pets allowed) and host reputation (host_is_superhost) affect listing prices. Determine which features add the most value and whether preferences differ across boroughs.

6. Timeline & Readiness Effect on Pricing [10 Marks]

Investigate how **booking readiness** (derive from availability_30: fully booked / limited availability / readily available) and **listing maturity** (years since first_review) influence prices in Manhattan and Brooklyn. Highlight trends and takeaways for guests planning short-term vs long-term stays.

7. Host ("Developer") Impact on Listings [15 Marks]

Analyse the impact of hosts on listing prices and features, using `calculated_host_listings_count` as a proxy for a host's "portfolio size" (the analog to a real-estate developer's project count). Determine if certain hosts (or larger-portfolio hosts) are associated with higher-end listings or better amenities.

Part III [10 Marks]

1. Code Quality [5 Marks]

Ensure the code is written in a vectorised way for data preprocessing using pandas, instead of writing individual for loops (wherever possible).

2. Report [5 Marks]

Based on your analyses from the previous questions, create a comprehensive report for both datasets. This should include data cleaning methods used as well as a small summary of findings in each dataset.